Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer with 2+ years building **C++/Java/Python applications**, full-stack platforms, relational data systems, and tested cross-platform software; experienced in OOP, data structures, algorithms, MVC, Agile/Scrum, CI, performance optimization, and secure delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python services and normalized relational schemas** for 500+ users, implementing row-level isolation, indexed queries, and optimized access patterns that reduced application dashboard latency from 1.2 seconds to 350 milliseconds.
- Implemented **validation, reconciliation, and failure-handling workflows** across 5,000+ monthly jobs, achieving 99%+ production reliability while improving diagnosis of malformed records, failed integrations, and recurring runtime defects.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **full-stack React and Flask features** using a Blueprint-based MVC architecture for authentication, uploads, processing, and history workflows, supporting 150+ users and 10,000+ daily requests in production.
- Developed **concurrent batch-processing components** with secure file ingestion, validation, persistent results, and fault-tolerant execution, processing 1,000+ files per batch while maintaining stable performance under peak application load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Implemented **data structures and planning algorithms** in Python across 100K+ trajectory records, integrating spatial-temporal features and reproducible simulations that improved long-horizon application performance 18% across benchmark scenarios.
- Assessed **performance parameters and failure patterns** across three architectures, documenting benchmark evidence and technical rationale that guided refactoring and improved closed-loop system stability 15% across simulation tests.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Designed **object-oriented API components** using Singleton and Factory patterns, SQL, caching, and AWS integrations, improving response times 30%, pipeline efficiency 68%, storage costs 15%, and maintainability across distributed workflows.
- Worked in **cross-functional Agile/Scrum delivery** with Microsoft and GitHub stakeholders, maintaining 20+ unit/integration suites and Jenkins CI across Windows, Linux, and macOS while reducing technical debt 20%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **full-stack React, Node.js, and Azure SQL features** across eight microservices for authentication, records, billing, permissions, and notifications, synchronizing 2M+ enterprise records through REST APIs and WebSockets.
- Implemented **product-security and testing practices** with JWT, role-based access, Jest, Selenium, Snyk, and dependency checks, then automated repeatable Docker deployments through Jenkins CI/CD, cutting release time 50%.

Skills

Languages: C++, Java, Python, JavaScript, TypeScript, SQL, Bash, HTML, CSS
Core Engineering: Data structures, algorithms, OOP, design patterns, performance optimization, refactoring
Architecture: MVC, microservices, REST APIs, FastAPI, Flask, Node.js, React.js, WebSockets, full-stack development
Data: PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, schema design, query tuning, validation
Quality & Delivery: Agile/Scrum, Git, Jenkins, CI/CD, PyTest, Mockito, Jest, Selenium, Docker, product security

Projects

CPU Scheduling Simulator Live | *C++, React.js, Algorithms*
- Implemented **six scheduling and optimization algorithms** with interactive controls, Gantt visualization, and dynamic performance metrics, testing 1,000+ process combinations while reducing average turnaround time 25% across benchmark scenarios.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered database application** supporting 10+ SQL and NoSQL systems, accelerating query authoring, schema exploration, and table management through a cross-platform Java backend and TypeScript interface with Docker workflows.

HMIS Live | *TypeScript, MERN, REST APIs, Sentry*
- Built a **secure full-stack management application** with TypeScript dashboards, REST integrations, authentication, role-based workflows, notifications, and Sentry monitoring, reducing record retrieval time 2x and administrative workload 30%.